

WIRELESS HEALTH AND SECURITY MONITOR That Does Not Breach Privacy

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Some patients and the elderly require continuous monitoring, but it cannot be done through CCTV cameras in their bathroom or even bedroom as it affects their privacy. The same applies to prisoners and some other cases. This project would help you design a device that lets you monitor the person's health data like respiration and the activity being done, such as sleeping, walking, reading, moving, whether present or running away, without the use of a camera.

Fig. 1 shows the author's prototype with the screen showing the activity being done. The components required to build this device are listed under the Bill of Material table.

Software

First you need to add the ESP board to Arduino IDE. To install the Arduino IDE, you can follow the instructions on ESP website. You need to install and add the library for radar interfacing, so download and add the library with following link:

https://wiki.seedstudio.com/Radar_MR24BSD1/

Then install the ESP Dash library to the host page having dashboard to show real-time data of the person's activity and vitals through Wi-Fi. You need to modify the library and do some settings to access the RX2 and

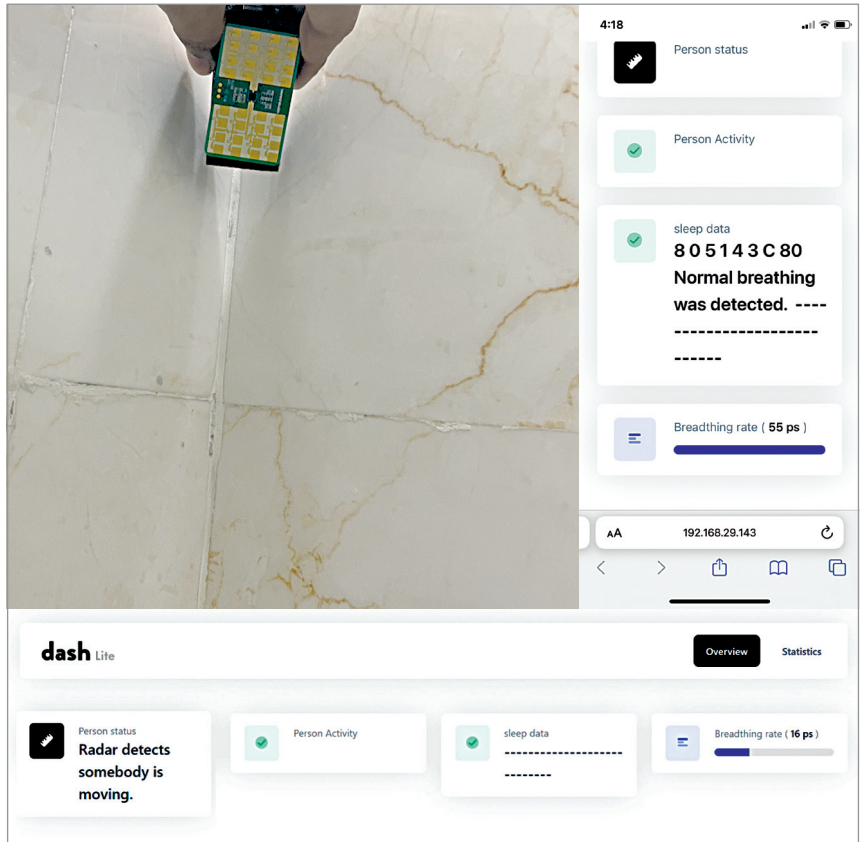


Fig. 1: Author's prototype

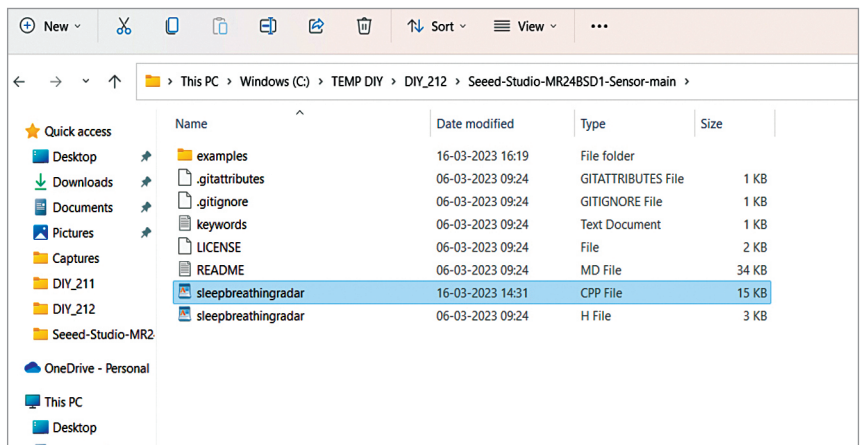


Fig. 2: Library file for modification

BILL OF MATERIAL

Components	Quantity	Description
MR24BSD1 sensor	1	24Hz mm-wave radar
ESP32 board	1	ESP 32 DEV board
Jumper wires	10	Female-female